

Human Embryology

HAWLER MEDICAL UNIVERSITY

COLLAGE OF MEDICINE

DEPARTMENT OF ANATOMY AND PATHOLOGY

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HUMAN EMBRYONIC DEVELOPMENT



FERTILIZED EGG



2 - CELL STAGE



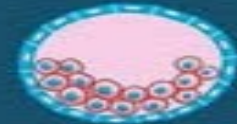
4 - CELL STAGE



8 - CELL STAGE



16 - CELL STAGE



BLASTOCYST



FOETUS
(5 WEEKS)



FOETUS
(10 WEEKS)



FOETUS
(20 WEEKS)



FOETUS
(40 WEEKS)



:Definition

Embryology is the branch of anatomy and developmental biology that studies the formation, growth, and maturation of the human from fertilization to the end of the embryonic and early .fetal periods

Embryology explains how normal anatomy forms and why congenital malformations occur, .making it essential for understanding pediatrics, obstetrics, surgery, radiology, and pathology

:Why embryology is important

:For medical students, embryology focuses on

Gamete formation (spermatogenesis and oogenesis) ·

Fertilization and implantation ·

Development of germ layers (ectoderm, mesoderm, endoderm) ·

Organogenesis (formation of organs and systems) ·

Congenital anomalies and their developmental basis ·

:What are the stages of development

Pre-embryonic stage (Weeks 1–2) .1

Embryonic stage (Weeks 3–8) (most critical stage) .2

Fetal stage (Week 9 → birth) .3

3 Stages of Fetal Development

The journey of life

GERMINAL STAGE (0–2 WEEKS)

- Conception occurs
- Zygote forms → blastocyst
- Travels to uterus
- Implants into uterine lining
- Hormonal changes begin



EMBRYONIC STAGE (3–8 WEEKS)

- Neural tube forms (brain/spine)
- Eyes, limbs, facial features develop
- Heartbeat by Week 6
- Major organs start forming



FETAL STAGE (9 WEEKS–BIRTH)

- Becomes fetus
- Sex assigned
- Hair, nails, eyelashes grow
- Limb movement begins



Pre-embryonic stage (Weeks 1–2) .1

:Main events

Fertilization ·

Cleavage → morula → blastocyst ·

Implantation ·

Formation of bilaminar disc (epiblast & hypoblast) ·

:Clinical relevance

.Failure here leads to implantation defects or early pregnancy loss

Embryonic stage (Weeks 3–8) (most .2 critical stage)

:Main events

Gastrulation → formation of three germ layers(Ectoderm, Mesoderm, • Endoderm)

Neurulation

Organogenesis •

Body folding •

:Clinical relevance

.Most congenital anomalies arise during this stage

Fetal stage (Week 9 → birth) .3

:Main events

Growth and maturation of organs ·

Functional development ·

Increase in size and weight ·

:Clinical relevance

.Problems mainly affect growth and function, not basic structure

:Types of cells involved in development

.Somatic cells .1

.Sex Cells (gametes) .2

:Somatic cells

.Somatic cells are all body cells except reproductive cells

:Key features

Form the tissues and organs of the body ·

Diploid ($2n = 46$ chromosomes) ·

Divide by mitosis ·

Responsible for growth, repair, and maintenance ·

Examples: skin cells, muscle cells, nerve cells

:Sex Cells (Gametes)

.Sex cells are reproductive cells involved in sexual reproduction

:Key features

Include sperm and ovum

Haploid ($n = 23$ chromosomes)

Formed by meiosis

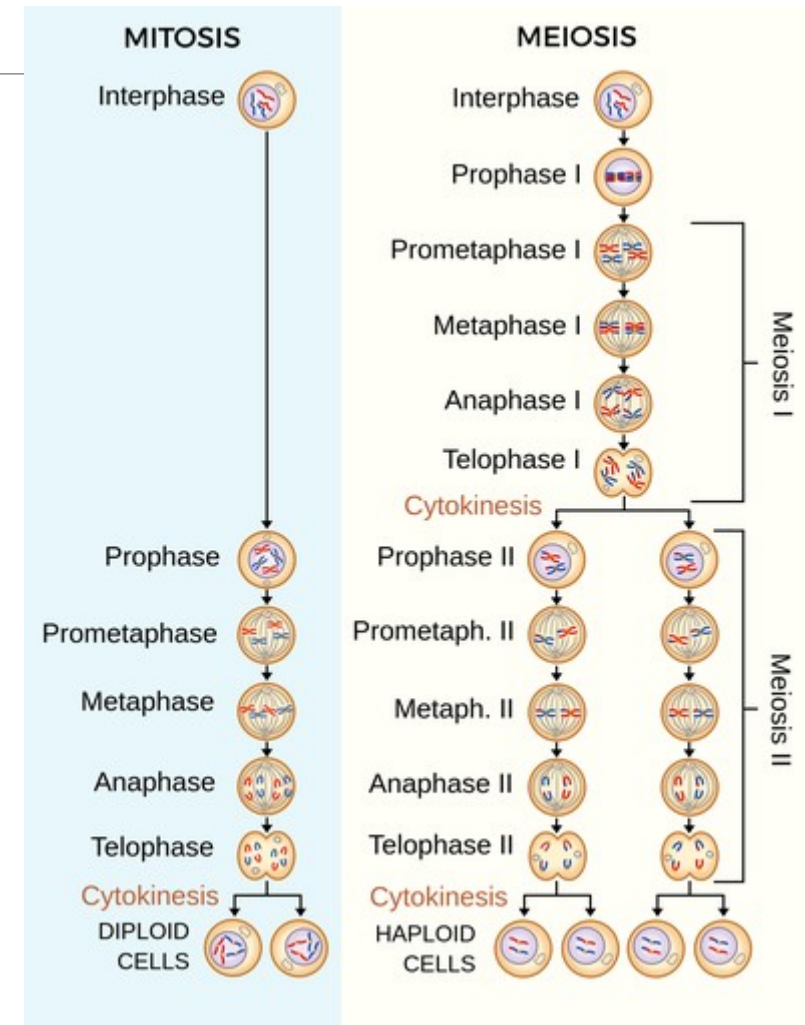
Fuse during fertilization to form a zygote

:Types of cell division

:There are two main types of cell division in humans

Mitosis .1

Meiosis .2



Mitosis .1

Occurs in: Somatic (body) cells

:Purpose

Growth •

Tissue repair •

Cell replacement •

:Key features

One division •

Produces 2 identical daughter cells •

Chromosome number maintained (diploid $\rightarrow$ diploid, $2n \rightarrow 2n$) •

Meiosis .2

Occurs in: Germ cells (in ovaries and testes)

:Purpose

Formation of gametes (sperm and ova) ·

Genetic variation ·

:Key features

Two divisions (Meiosis I & II) ·

Produces 4 non-identical daughter cells ·

Chromosome number reduced (diploid $\rightarrow$ haploid, $2n \rightarrow n$) ·

Special processes:

- Crossing over
- Independent assortment

:Gametogenesis

Gametogenesis is the process by which haploid sex cells (gametes) are formed from diploid germ cells through meiosis

:It includes

Spermatogenesis (male)

Oogenesis (female)

Spermatogenesis

Spermatogenesis is the process of formation of male gametes (spermatozoa) in the seminiferous tubules of the testes

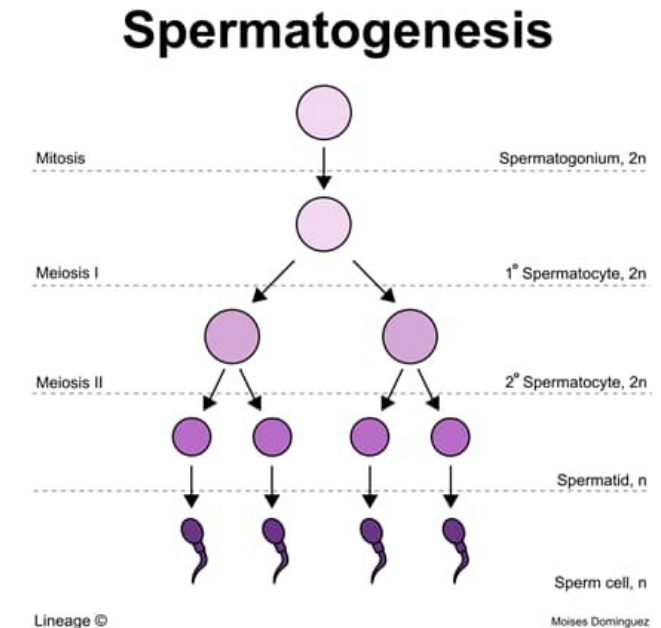
:Key points

Begins at puberty

Continuous process

One primary spermatocyte → 4 functional sperm

Produces small, motile cells



Oogenesis

.Oogenesis is the process of formation of female gametes (ova) in the ovaries

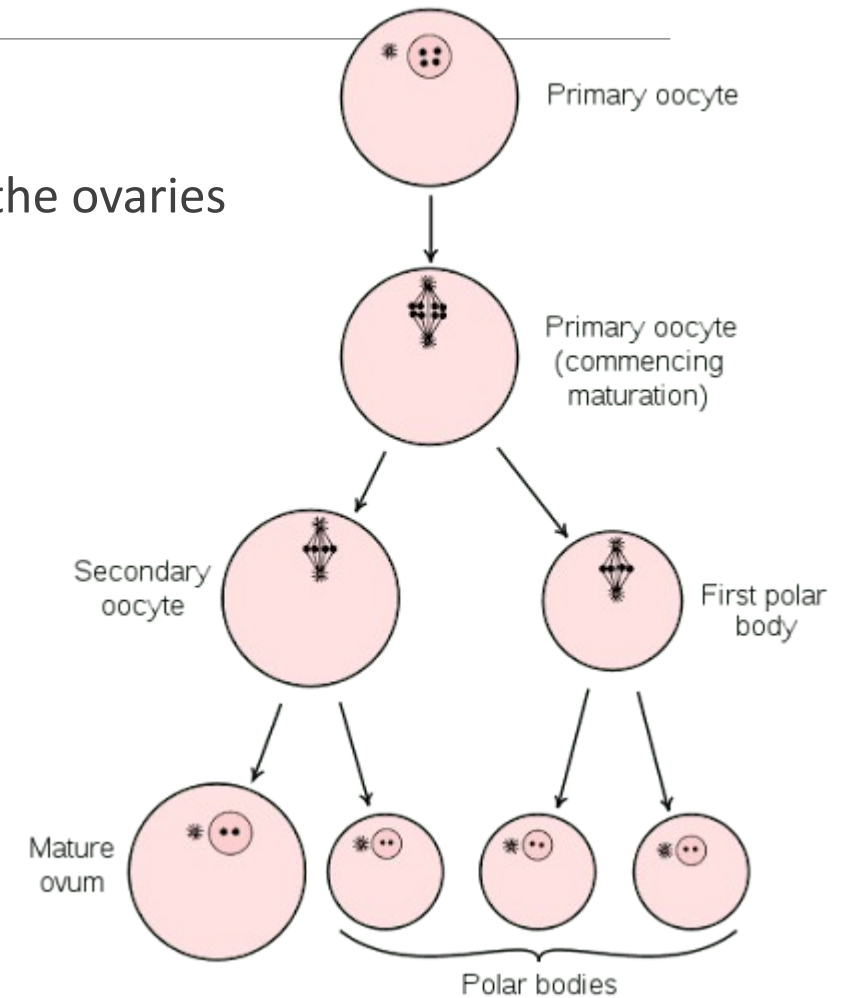
:Key points

Begins during fetal life

Interrupted at prophase I and metaphase II

Completed only after fertilization

One primary oocyte → 1 functional ovum + polar bodies



Fertilization

Fertilization is the process by which a male gamete (sperm) fuses with a female gamete (ovum) to form a zygote

:Key points

Normally occurs in the ampulla of the uterine tube

Restores the diploid chromosome number ($2n = 46$)

Determines genetic sex

Initiates embryonic development

FERTILIZATION AND IMPLANTATION

